

values can be achieved by using more advanced PV devices and improved TE materials, with a potential to reach 50% total efficiency.

Hybrid PV and solar thermal system

A technique that combines PV and solar thermal systems to efficiently convert solar radiation to electricity for immediate use and store the remaining inexpensive thermal energy (not utilised by PV) to convert to electricity on demand has been developed recently. It is called hybrid electric and thermal solar (HEATS) system. The prototype performs at 26.8% (with a potential to achieve 35.2%) solar to electricity efficiency and 81% dispatchability of electrical energy from thermal energy at an operating temperature of 775K using silicon PV cell (gallium arsenide PV cell can also be used).

HEATS contains a PV module as well as a thermal absorber to utilise the best of both. In this system photons in the PV band are directed to the PV cells whereas these are most efficiently converted into electricity. Low-energy photons (long wavelengths) that cannot be converted by PV cells and high-energy photons (short wavelengths), which would be converted inefficiently, are directed to the thermal absorber instead of being wasted.

This technique improves overall system efficiency and provides additional thermal energy, which can be stored at low cost to be used for electricity generation or for heating on demand. Dispatchability is the ratio of electricity generated from

heat engine and total electricity generated by both heat engine and the PV modules. The arrangement of HEATS is shown in Fig. 16.

Fig. 16(A) shows the receiver concept with a cutaway section to show its internal structure. The HEATS receiver is used in conjunction with a solar concentrator, like a parabolic trough or a linear Fresnel reflector, to increase the intensity of solar radiation. The stacked structure consists of a glass protective cover, a thermally insulating transparent aerogel layer, the spectrally selective light pipe (SSLP), followed by another insulating transparent aerogel layer, and finally the PV module.

The SSLP structure consists of a series of parallel fins attached to the heat collection pipes carrying a heat transfer fluid like Therminol VP-1, which collects the thermal energy absorbed by the SSLP. The SSLP structure is formed from parallel fins made of a thermally conduc-

tive copper sheet substrate coated with a spectrally selective material in multilayers.

The SSLP absorbs high and low energy photons as thermal energy, while directing the mid energy photons to the PV module down below as shown in Fig. 16(B). Transparent aerogel on either side allows the light to pass through but does not allow the heat to escape as it is an excellent insulator.

The aerogel layers serve to thermally insulate the SSLP from the PV module to keep the latter cool. It ensures that the SSLP can be operated at a high temperature without heat loss while the PV module and the glass cover remain at a safe low operating temperature. The transparent silica aerogels used have high solar transmittance of about 96% and low thermal conductivity of about 0.055W/m.K. As aerogels are not very strong, it is ensured that no load is transferred to it.

The advantages of hybrid PV-thermal system are mainly: (a) both solar thermal and PV cells can be housed in the same module and operated simultaneously, (b) the solar energy that would have been lost otherwise, if single PV module was used, can be recuperated usefully, and (c) the amount of energy that is generated per unit area by this tandem system is more and the payback period is less due to more energy extraction. **EFY**

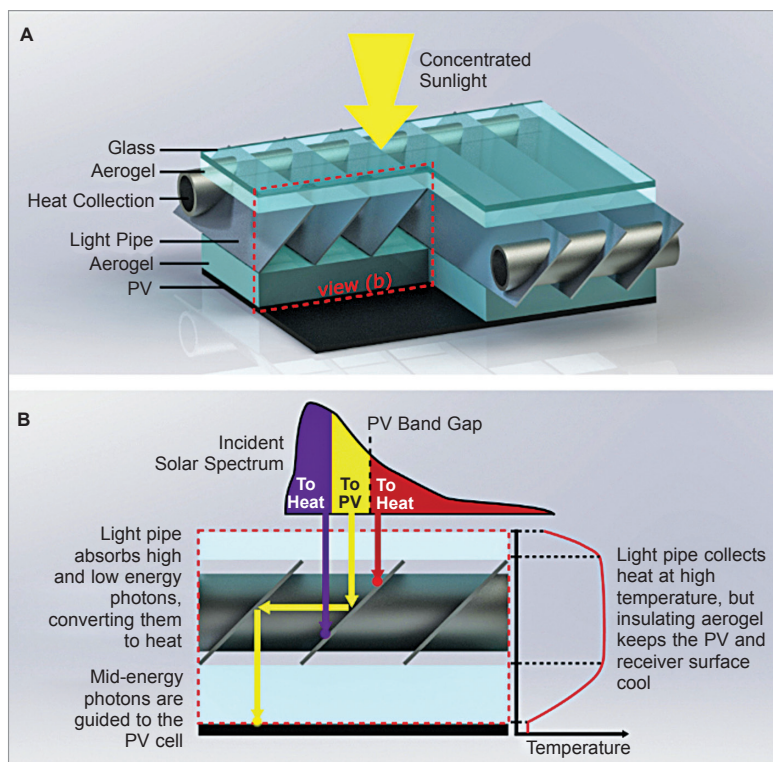


Fig. 16: Hybrid PV and solar thermal system